Loan Default Prediction

Home Credit Default Risk

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Applicant data and credit history

Source	Applicant-level input
Application table	Income, credit amount, employment, scores
Historical tables (optional)	Mean, max, min and record count
Bureau balance	Status counts linked through bureau ID

307,511 applicants; 8.07% positive outcomes; 126 input features

Feature engineering and preprocessing

Credit / income

Same loan amount, different repayment pressure

365243 employment sentinel
Missing value + separate indicator

Numeric variables
Median imputation and scaling

Categorical variables
Missing category and one-hot encoding

Each training fold fits its own preprocessing pipeline

Model comparison

Model	Mechanism	Tuned setting
Logistic regression	Linear score, logistic link	Regularization C
Random forest	Average randomized trees	Maximum depth
LightGBM	Sequential loss reduction	Leaves and L2 penalty

Selection criterion: highest development CV AUC

Validation and threshold selection

80% development

Stratified CV (default: 3 folds)
Grid search by AUC
OOF threshold by maximum F1

20% holdout

Final evaluation
After selection

Tuned CV / OOF can be optimistic. Holdout never selects the model.

Performance and error tradeoffs

Model	CV AUC	Holdout AUC
Logistic regression	0.7492	0.7521
Random forest	0.7401	0.7443
LightGBM	0.7607	0.7649

Selected model: LightGBM **Threshold: 0.1563**

Precision: 25.67% Recall: 41.41% F1: 0.3169

307,511 applicants; 126 input features; historical-table use unconfirmed

Model interpretation and final predictions

Permutation importance

Shuffle one feature
Measure the AUC decrease
Computed on development data

Final submission

Refit on all labelled data
Predict unlabelled applicants
Export default probabilities

SHAP explains LightGBM. Importance describes association, not causation.

Limitations and next steps

Timing of historical inputs needs a field-level audit

F1 threshold does not optimize lending profit

Next: probability calibration and a cost-based policy

github.com/Xyphorial/home-credit-default-prediction